WebGen-R1: Incentivizing Large Language Models to Generate Functional and Aesthetic Websites with Reinforcement Learning

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Research Questions

- How can reinforcement learning be designed to optimize both functional correctness and aesthetic quality in end-to-end multi-page website generation?
- What reward mechanisms can reliably evaluate website generation without prohibitively expensive GUI agent exploration?
- Can small open-source LLMs (7B) be trained to match or exceed much larger models (72B+) in project-level website generation tasks?

Core Idea

Website generation requires both functional correctness and aesthetic appeal, but existing approaches either simplify to single-page sites or use expensive multi-agent frameworks. The key insight is to constrain the generation space using a pre-validated React template scaffold, then use a cascaded reward system that cheaply filters out broken code before applying expensive visual evaluation. This combines structural guarantees (template ensures buildability), execution verification (automated testing), and aesthetic assessment (VLM-based scoring) in a computationally feasible RL framework.

Key Formula

Cascaded Multimodal Reward

`R(y) = \begin{cases} \psi_{static}(W_{gen}) & \text{if } I_{static}(W_{gen}) = 0 \\ \psi_{build}(\Lambda_{runtime}) & \text{if build fails} \\ R_{dense}(y, o, x) & \text{otherwise} \end{cases}`

Hierarchical reward function where R(y) is total reward, ψ_static penalizes static compliance failures, ψ_build penalizes build failures, and R_dense provides dense rewards only for successfully rendered projects. This avoids expensive VLM evaluation on broken code.

Dense Reward Components

$$R_{dense} = s_{vis} + \gamma \cdot s_{func} + \lambda \cdot s_{cot}$$

For successfully rendered websites, combines s_vis (VLM aesthetic score), s_func (functional integrity from error logs), and s_cot (reasoning format reward). Weights γ and λ balance functional vs aesthetic objectives.

Key Figure

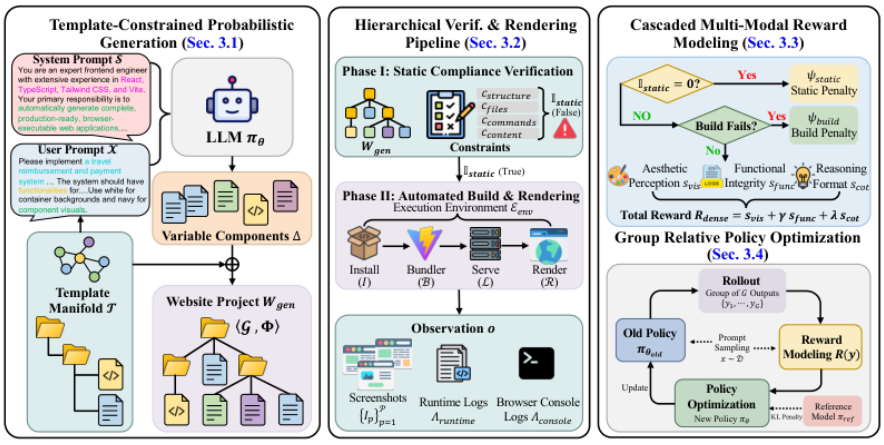

Figure 1: Overview of WebGen-R1, a reinforcement learning framework for functional and aesthetic website generation. Given a system prompt, a user specification, and a template manifold, the policy π_θ generates only the variable project components, which are then injected into predefined scaffold slots to form a complete website project. The generated project is evaluated by a hierarchical verification and rendering pipeline. In Phase I, static compliance verification checks structural, file-level, command-level, and content-level constraints and terminates invalid samples early. In Phase II, statically valid projects proceed through dependency installation, bundling, local serving, and browser-based rendering, producing screenshots together with runtime and console logs. Based on these observations, WebGen-R1 uses a cascaded multimodal reward that assigns scalar rewards to static or execution failures and otherwise combines a VLM-based aesthetic perception score, a functional integrity score derived from execution logs, and a reasoning format reward. The resulting reward is used to optimize the model with GRPO. This design combines template-level structural constraints, execution-grounded functional reward supervision, and visually informed aesthetic reward supervision, enabling efficient end-to-end RL for high-quality multi-page website generation.

Figure 1: Shows the complete WebGen-R1 framework architecture including template-constrained generation, hierarchical verification pipeline, cascaded reward modeling, and GRPO optimization in a single comprehensive overview.

Limitations

- Relies on pre-validated React template which may not generalize to other web frameworks or architectures
- VLM-based aesthetic evaluation may not capture all aspects of user experience or design quality
- Static compliance verification rules are manually designed and may miss edge cases
- Evaluation limited to specific benchmark datasets that may not represent full diversity of real-world websites